Examiner only

12. (a) **Y** is a halogenocompound in which each molecule contains one atom of chlorine, bromine or iodine.

(i) Describe a chemical test to determine which halogen is present in **Y**. [3]

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(ii) **Y** contains four carbon atoms in each molecule. The percentage by mass of halogen present in **Y** is less than 40%.

Identify **Y**. Explain how you reached your conclusion. [2]

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(b) (i) Halogenoalkanes react with aqueous sodium hydroxide.

Draw the mechanism to show the reaction of 1-chloropropane with aqueous sodium hydroxide.

You should include all charges, partial charges and lone pairs, and curly arrows to show electron movement. [4]

(ii) Name the type of reaction shown in part (i). [1]

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(c) Halogenoalkanes can also take part in an elimination reaction.

2-Chloropentane undergoes elimination in a similar way to 1-chloropropane.

(i) Give the reagent and conditions needed for 2-chloropentane to undergo elimination. [1]

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(ii) When 2-chloropentane undergoes elimination, two structural isomers are formed.

Give the structures of these **two** isomers. [2]

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